

LecW04A Monday: Asynchronous. Recording's link will be in Brightspace. I'll be traveling for a conference.

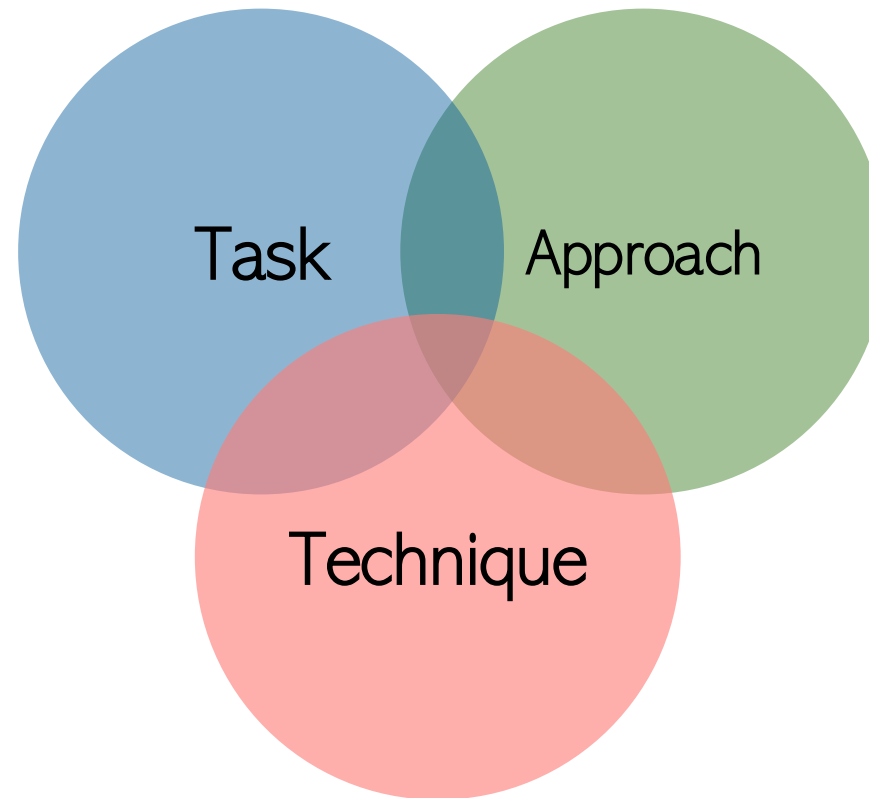


An image of the core region of a supermassive black hole, processed from a sparse array of radio telescopes
https://en.wikipedia.org/wiki/Black_hole

Outline: W02B: Topics Start

Classification

Regression
Clustering
Ranking (Recommendation)
Vector Representation and
Dimensionality Reduction
Language Modeling



Full/Self/Semi-Supervised Learning

Unsupervised Learning
Reinforcement Learning

k-Nearest Neighbor, Naive Bayes, Logistic Regression
Artificial Neural Networks, Graph Neural Network
Support Vector Machines
Principal Component Analysis
K-Means

4 Outline: W03A: Topics Start

- I) Task: Classification
- II) Supervision: Fully Supervised
- III) Technique: k -Nearest Neighbor
- III) Evaluation

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k -Nearest Neighbor: Natural Language

"We become who we hang out with" - Napoleon Hill

"If you hang around 5 confident/intelligent/millionaires/idiots, you will be the 6th – Denzel Washington



Oliver Napoleon Hill (1883 –1970)

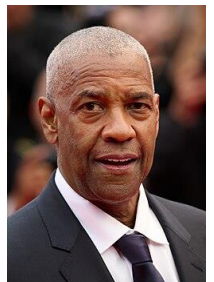
American self-help author and **conman**.

Known for his book *Think and Grow Rich* (1937), the best-selling self-help books of all time.

Most of his books are promoted as expounding principles to achieve "success".

A controversial figure, **history of criminal charges and fraud accusations**, historians also doubt many of his claims.

Two Academy Awards, an Actor Award, two Golden Globes, two Silver Bears and a Tony Award as well as nominations for a Grammy Award and two Primetime Emmy Awards.



Denzel Hayes Washington Jr. (1954-Present)

American actor and producer.

Known for his dramatic roles on stage and screen.

6 Outline: W03A: Topics Start

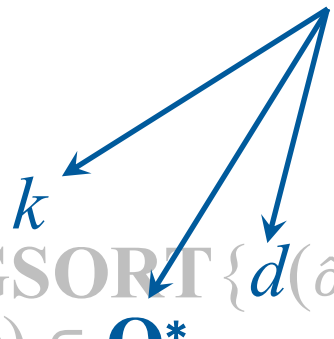
- I) Task: Classification
- II) Supervision: Fully Supervised
- III) Technique: k -Nearest Neighbor
Let's zoom in ... Hyperparameters
- III) Evaluation



Tug of war (also known as tug o' war, tug-a-war, tug war, rope war, rope pulling, or tugging war)

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k-Nearest Neighbor: Let's zoom in ...

$$f(\hat{o} : \mathbf{O}^*, k, d) = \underset{c_i \in \mathbf{C}}{\text{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \text{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}} \mathbf{1} [c_i = c] \right\}$$


No internal memory (parameters) → *non*-Parametric (no summarization)

But it has Hyperparameters

→ Compile-time parameters

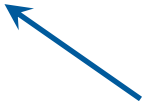
Form the learning algorithm architecture (structure)

Are “*tuned*” during design to find the best architecture

Remain constant after deploy for production

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k -Nearest Neighbor: Efficacy vs. Efficiency Trade-Off

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k 1 [c_i = c] \right\}$$


Scaling labeled dataset:

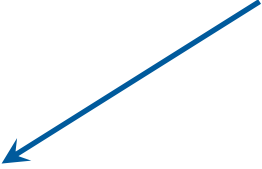
→ Small → Fast but **Poor**

→ Large → **Slow** but Accurate

Labeled set of observations (training set) is commonly not considered a hyperparameter. Why?

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k -Nearest Neighbor: Variance vs. Bias Trade-Off

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k 1 [c_i = c] \right\}$$


k , number of nearest neighbor

→ Small like $k = 1$

Liberal: decides based on whatever sits next to $\hat{o}$

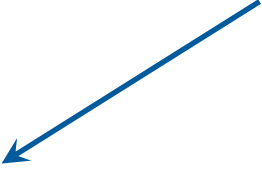
Unbiased (Low Bias): not impose a strong global preference

Locally Sensitive

High Variance

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k -Nearest Neighbor: Variance vs. Bias Trade-Off

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k 1 [c_i = c] \right\}$$


k , number of nearest neighbor

→ Large like $k = |\mathbf{O}^*|$

Conservative: considers a broader neighborhood

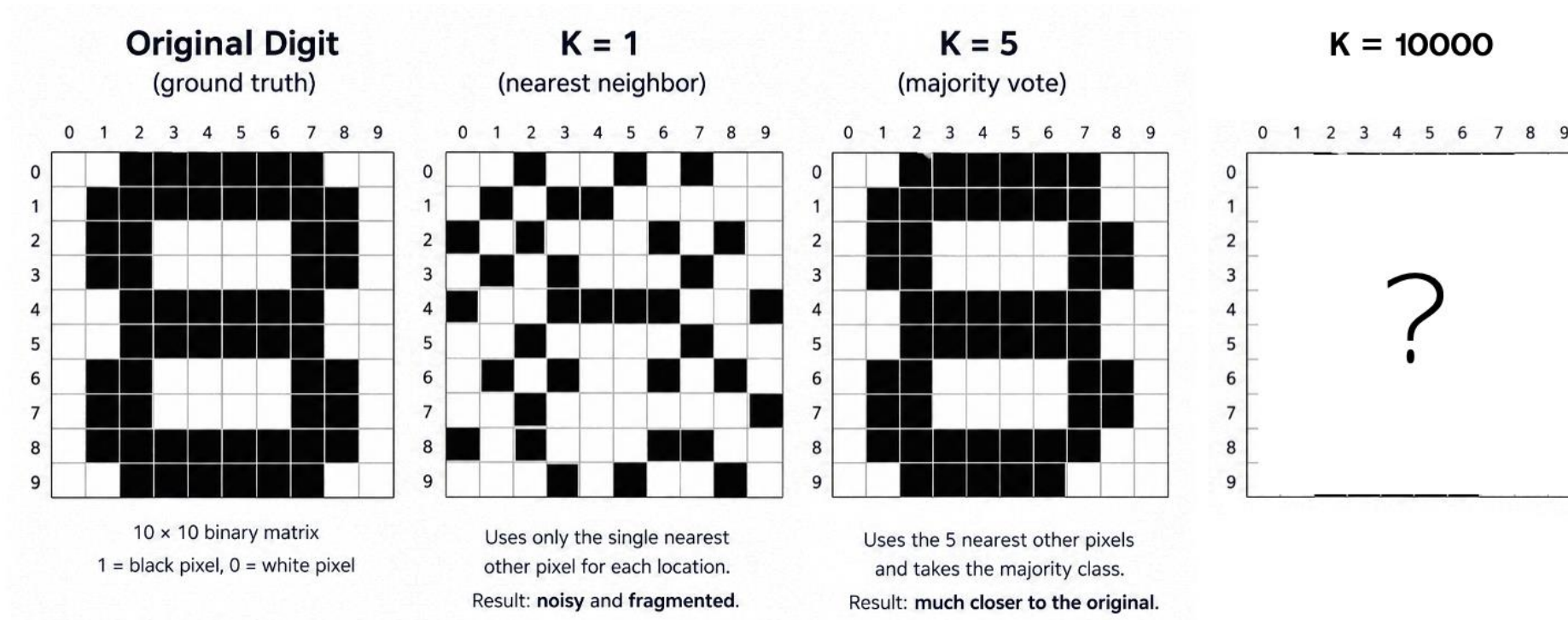
Low Variance

Biased (High Bias): Blind, favors the global over local pattern

Globally Conforming, Another Brick in the Wall :|

k -Nearest Neighbor: Inference on Training Set

A good learner (technique) can easily learn the training set.
A student whose exam contains only the lecture contents.



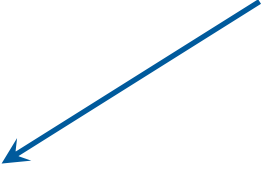
We make a copy of an image, then restore every pixel of it by its surrounding ...



Daniel Edmond Sullivan Jr
Ocean Photographer of the Year
<https://www.nationalgeographic.pt/fotografia/imagens-que-mostram-beleza-oculta-mares-e-oceanos-concurso-ocean-photographer-of-the-year> 7414

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 r -Nearest Neighbor: Neighbors vs. Neighborhood

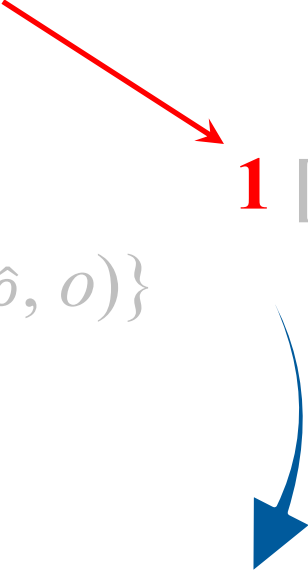
$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}} \underset{d \leq r}{1} [c_i = c] \right\}$$


$k \in \mathbb{N}^+$ number of nearest neighbors $\rightarrow r \in \mathbb{R}^{\geq 0}$ radius of nearest neighborhood

- $\rightarrow$ Small, like r close to 0 $\rightarrow$ Empty neighborhood, cannot decide.
- $\rightarrow$ Large, like $r = +\infty \rightarrow$ Very crowded neighborhood, conformity.

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 k -Nearest Neighbor: Weighted (“Attentive”)

$$f(\hat{o} : \mathbf{O}^*) = \operatorname{ARGMAX}_{c_i \in \mathbf{C}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k \mathbf{1} [c_i = c] \right\}$$


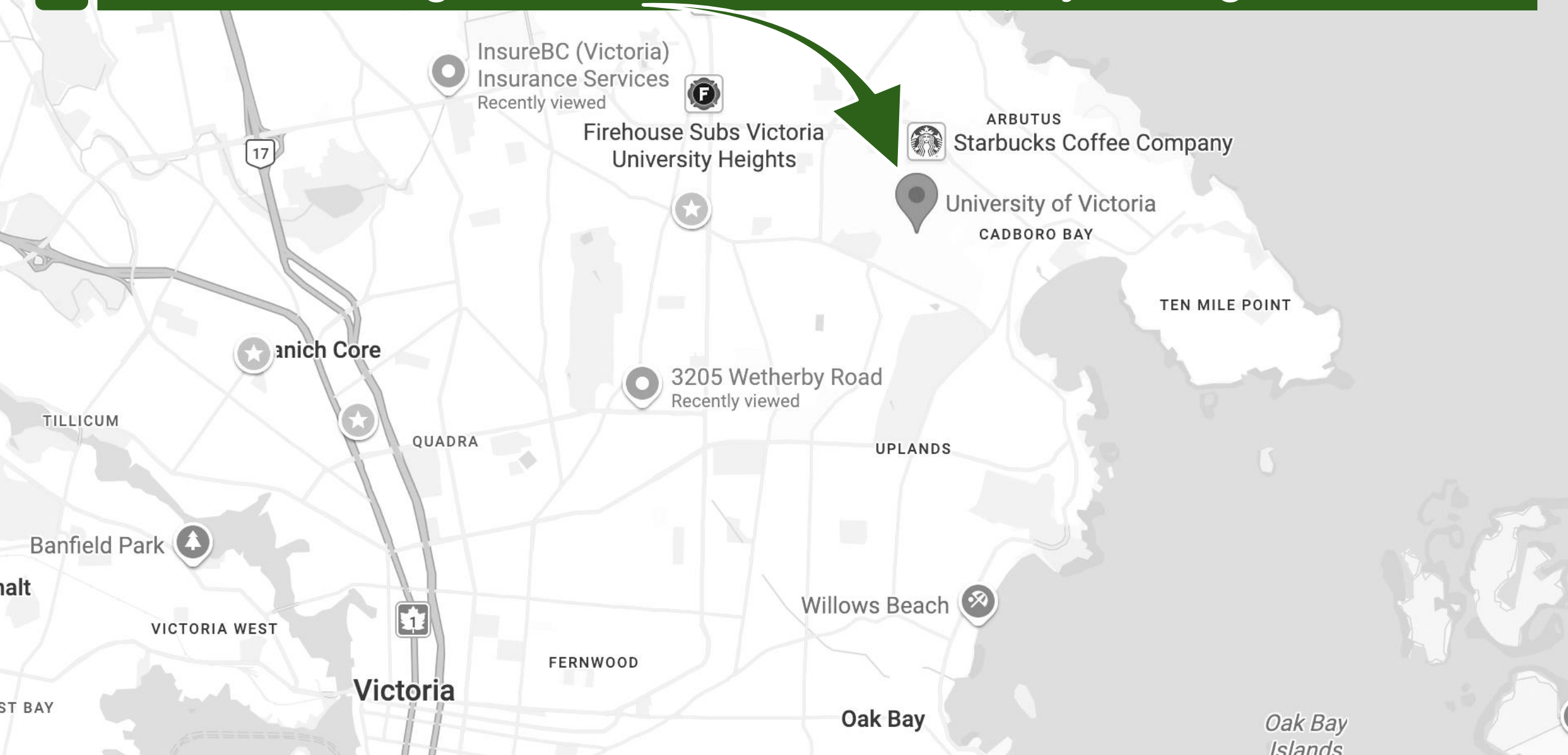
$$f(\hat{o} : \mathbf{O}^*) = \operatorname{ARGMAX}_{c_i \in \mathbf{C}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k \frac{1}{d(\hat{o}, o)} \times [c_i = c] \right\}$$

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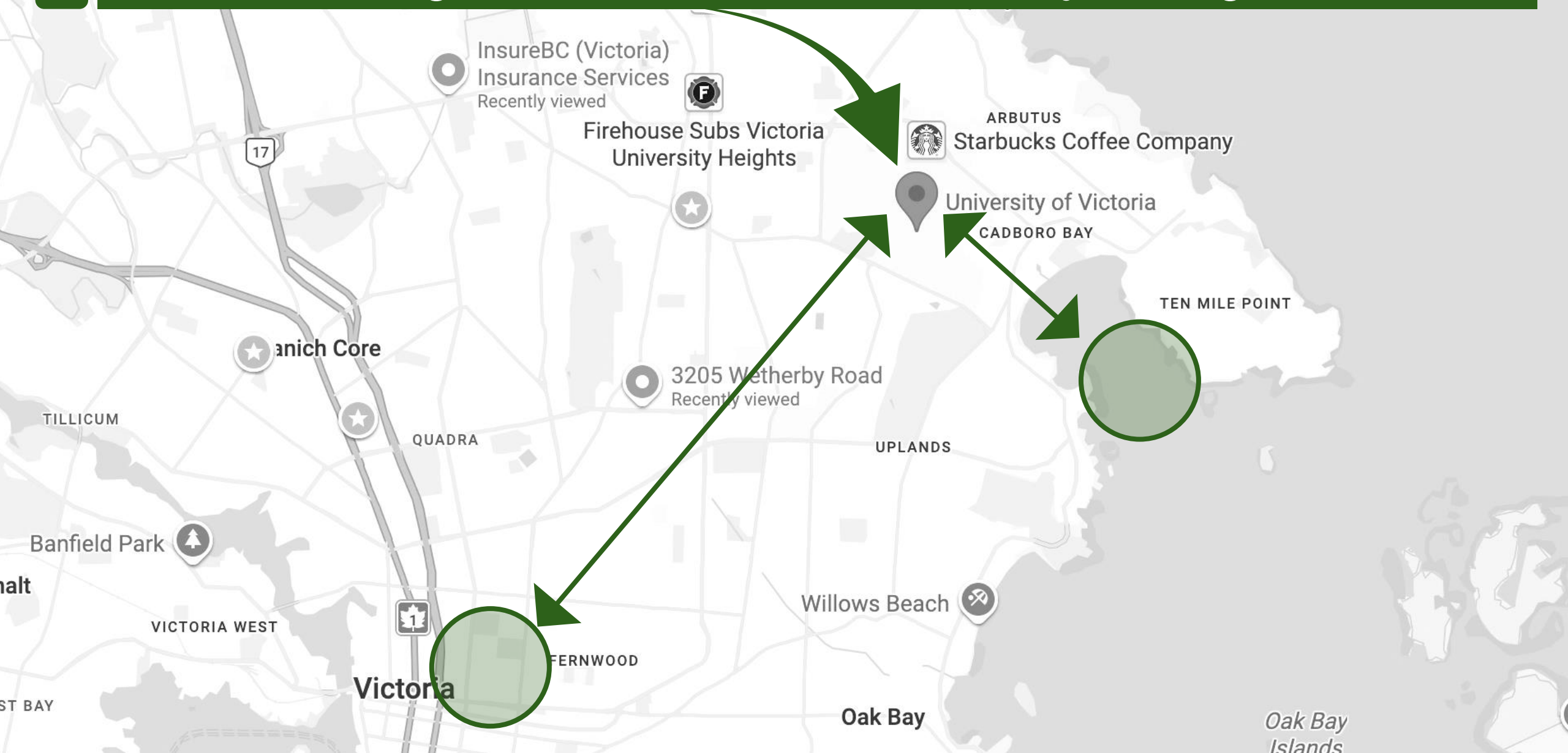
 k -Nearest Neighbor: Distance Function

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k 1 [c_i = c] \right\}$$

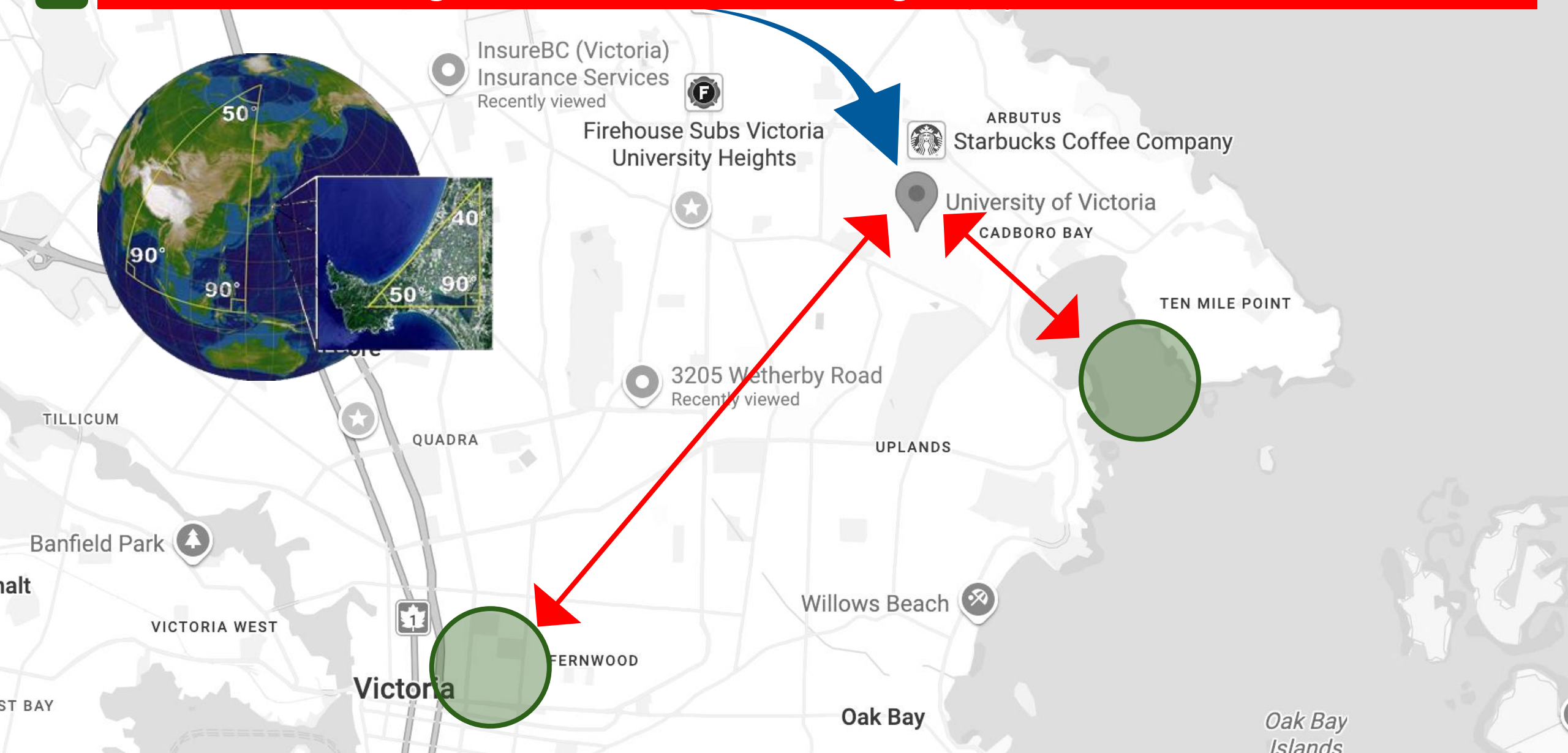

k -Nearest Neighbor: Traffic at UVic? Heavy vs. Light



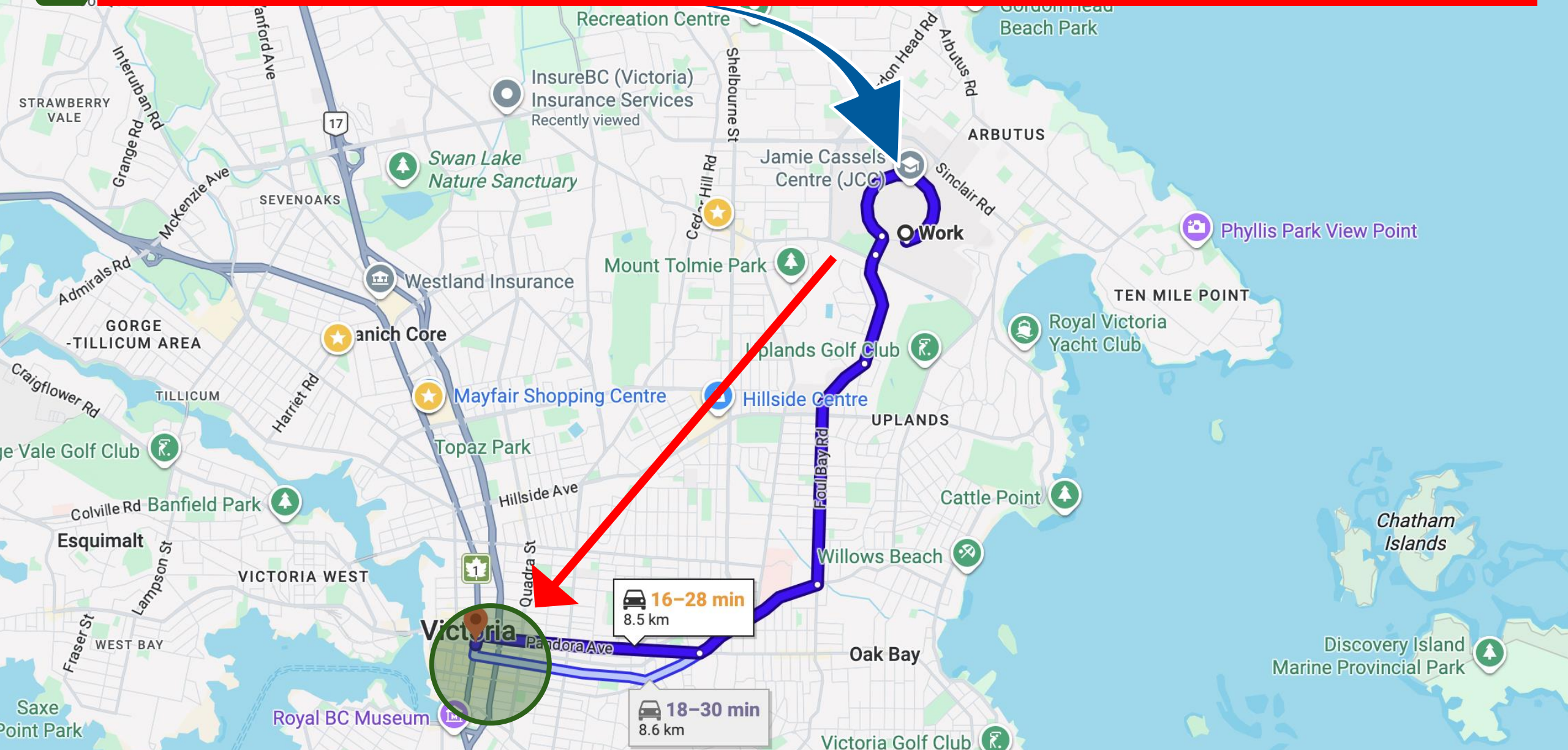
k -Nearest Neighbor: Traffic at UVic? Heavy vs. Light



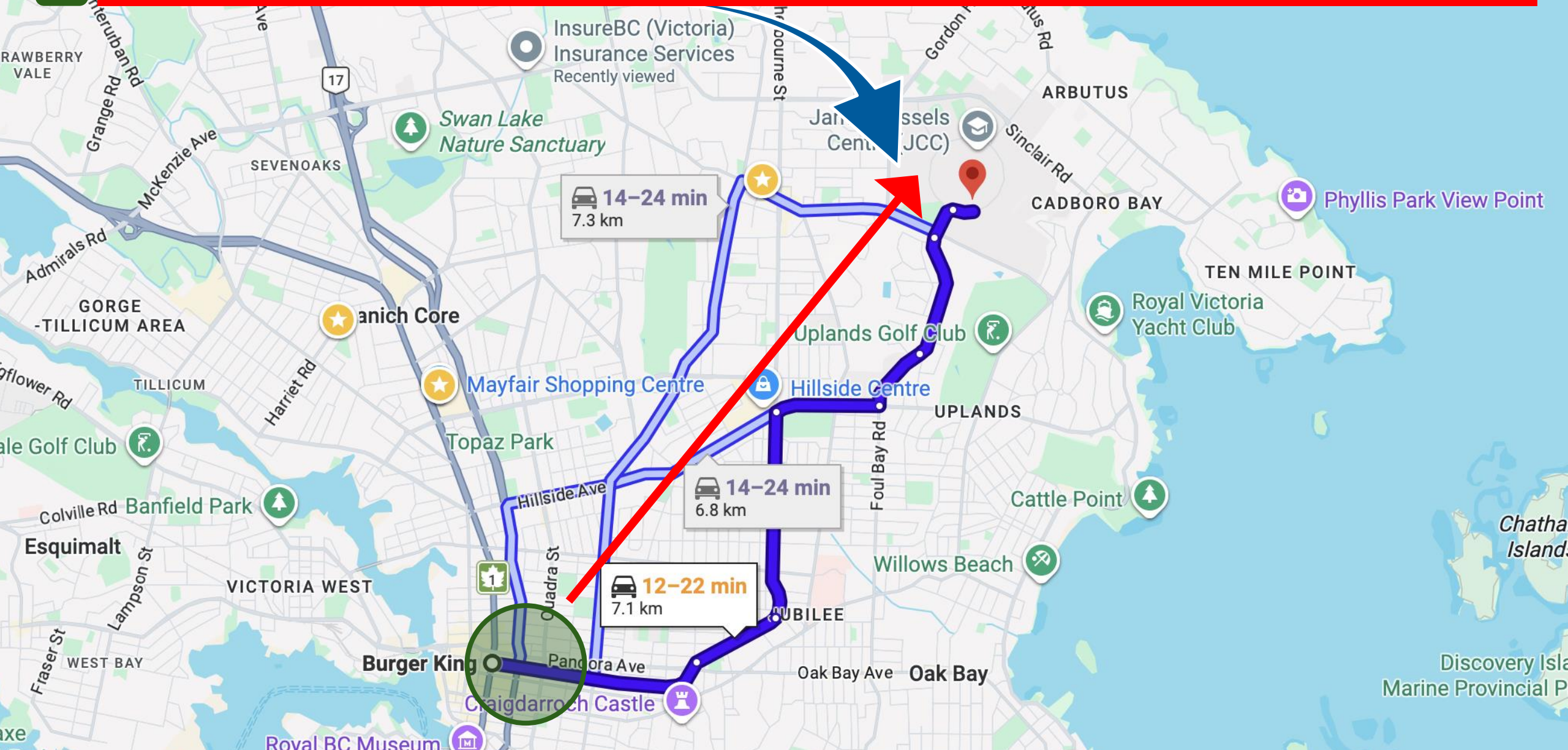
k -Nearest Neighbor: Math Challenges



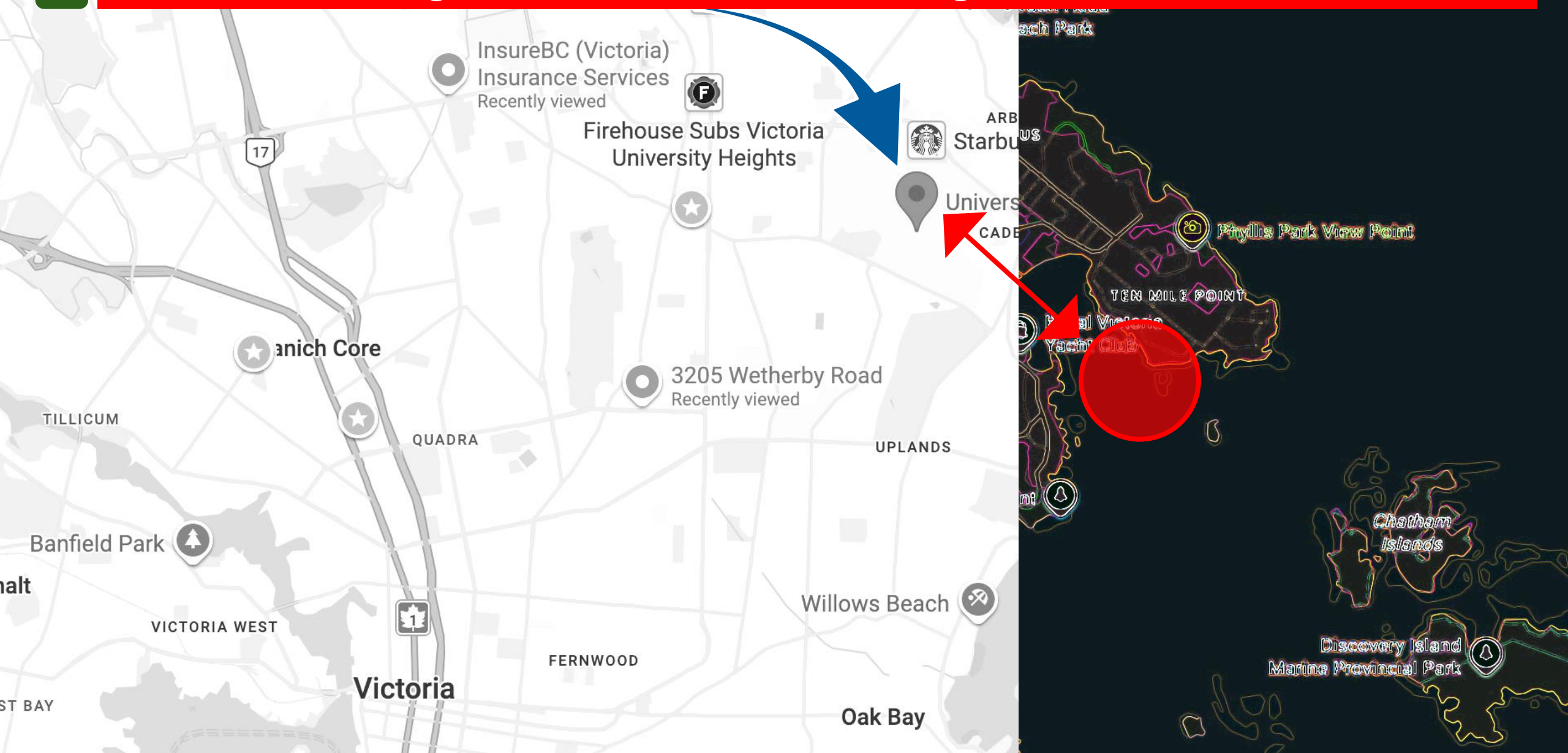
k -Nearest Neighbor: Domain Challenges



k -Nearest Neighbor: Domain Challenges



k -Nearest Neighbor: Domain Challenges



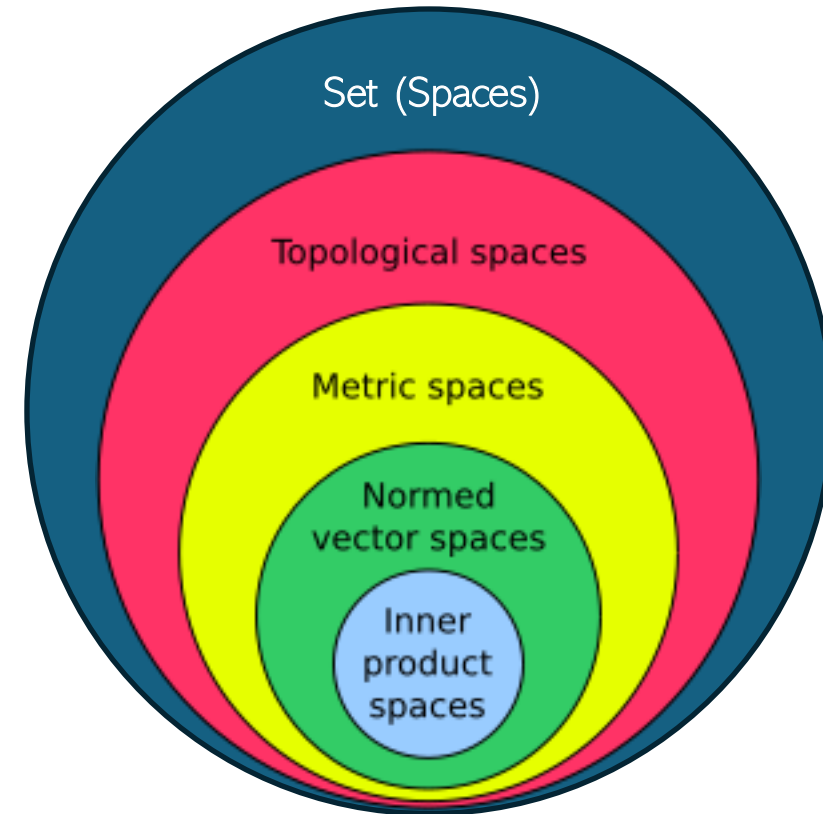
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k -Nearest Neighbor: Space Hierarchy

We need a space that covers observations for defining distance function.

More restrictions
More assumptions
More structure
More idealized
Less realized
Less general
Easier mathematics
Easier to work with

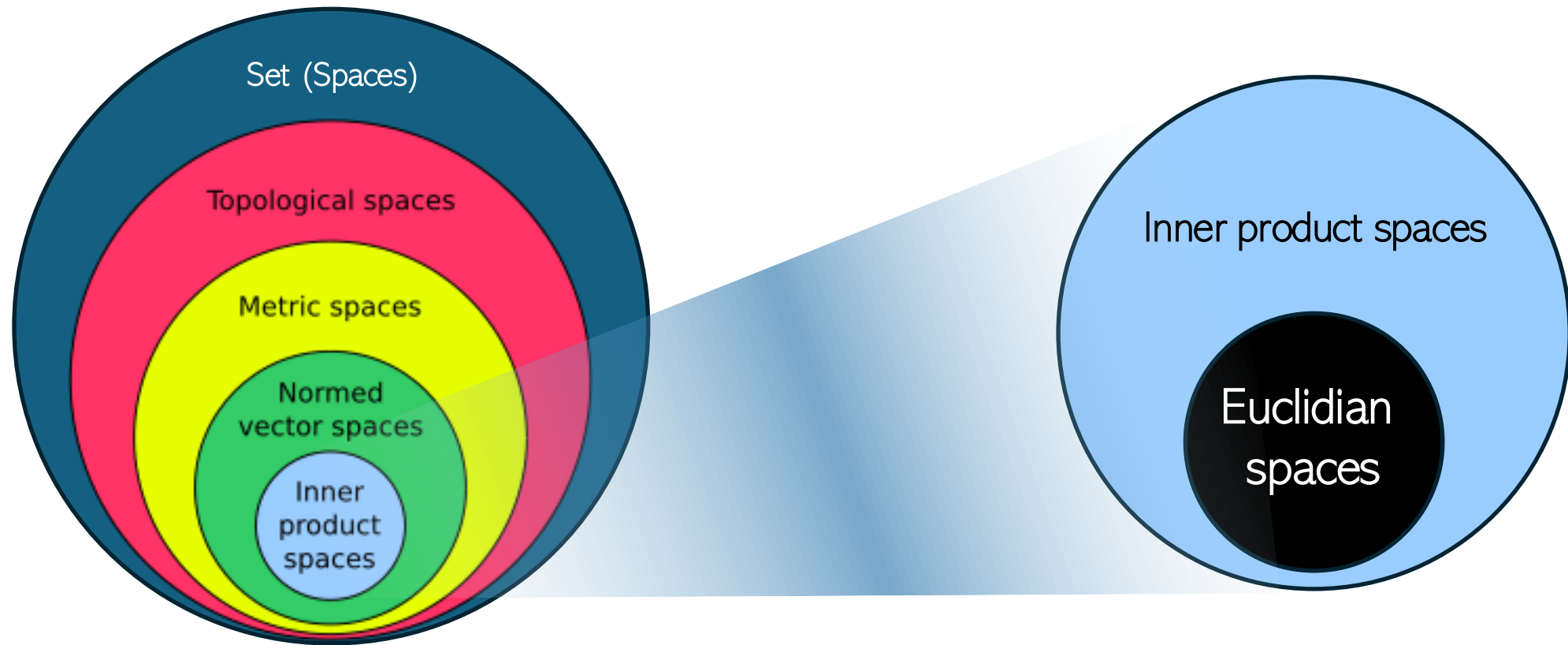
“give up generality to gain tractability”



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k -Nearest Neighbor: Space Hierarchy

We need a space that covers observations for defining distance function.



“nice geometric space”

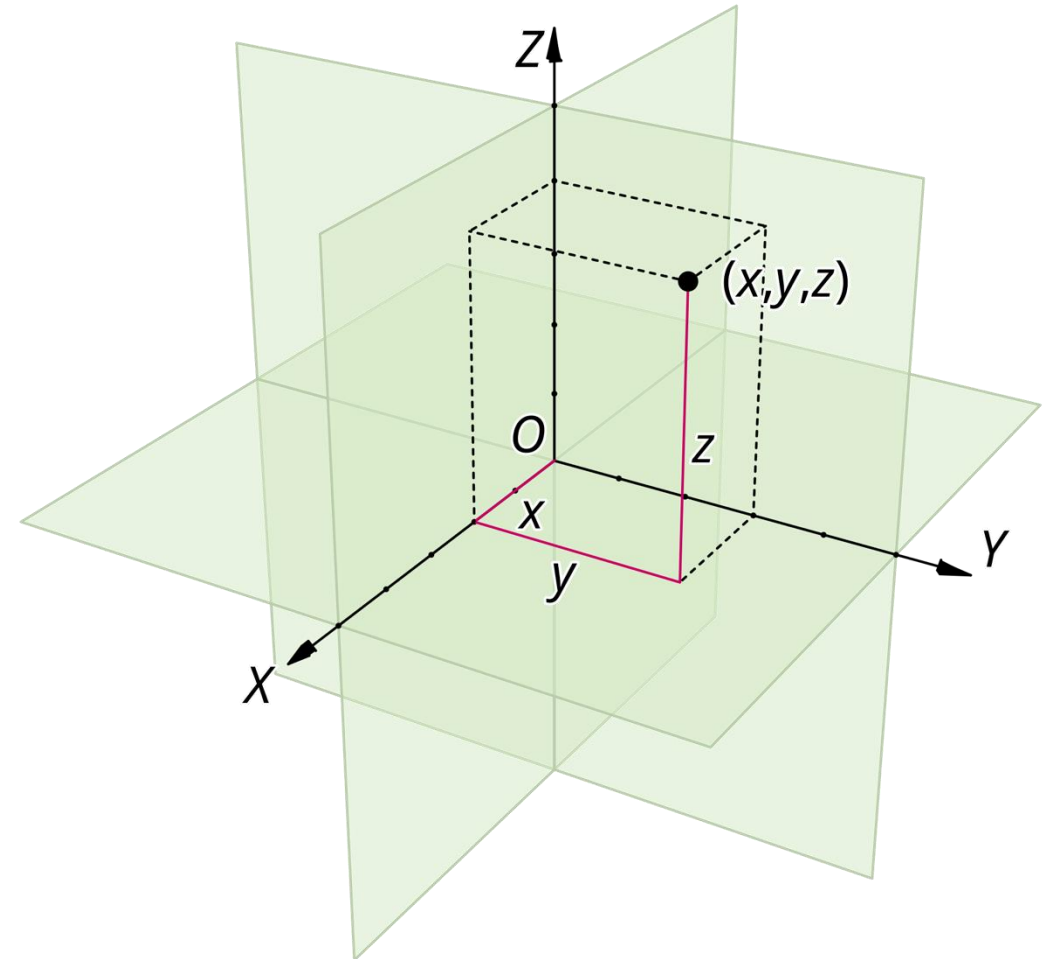
yoo-KLID-ee-uhn

k -Nearest Neighbor: Euclidian Space



Detail of Raphael's impression of Euclid, teaching students in *The School of Athens* (1509–1511)

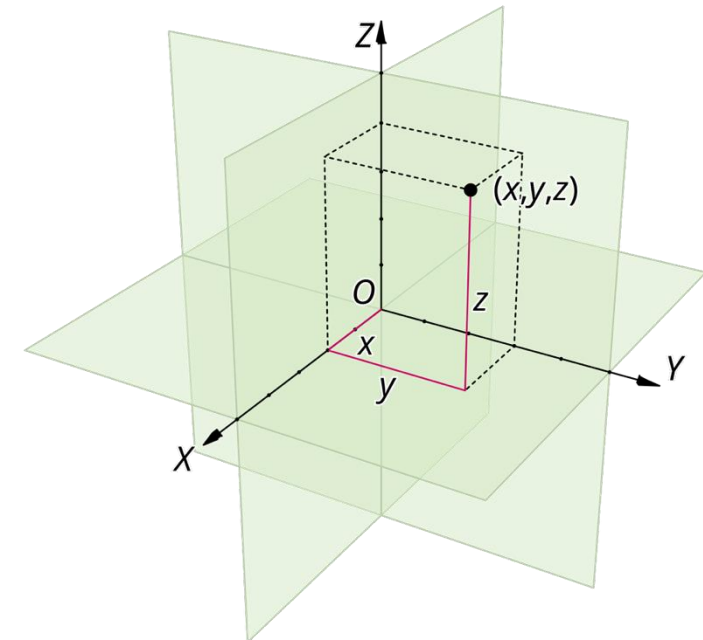
Euclid (/ˈjuːklɪd/; Ancient Greek: Εὐκλείδης; fl. 300 BC)



k -Nearest Neighbor: Euclidian Space

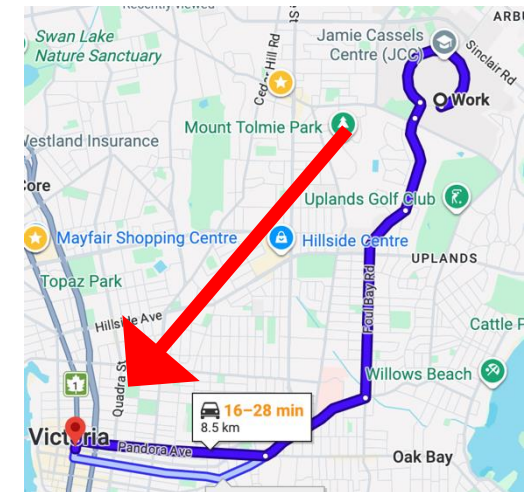
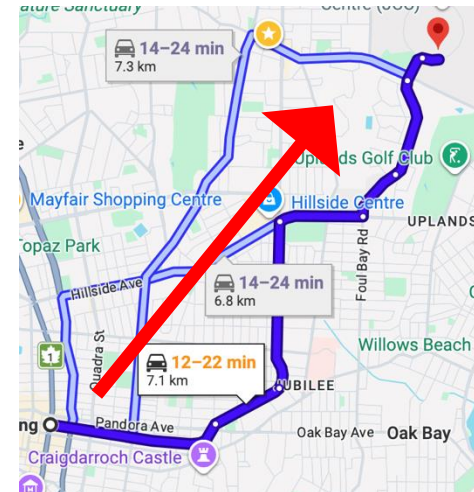
If $\mathbf{p}$ and $\mathbf{q}$ are two “points” (pls don't use the term “vector” for now):

1. $d(\mathbf{p}, \mathbf{q}) \geq 0.0$ (no negative distances).
2. $d(\mathbf{p}, \mathbf{q}) = 0.0 \Leftrightarrow_{\text{iff}} \mathbf{p} = \mathbf{q}$.
3. $d(\mathbf{p}, \mathbf{q}) = d(\mathbf{q}, \mathbf{p})$ (symmetric).
4. $d(\mathbf{p}, \mathbf{q}) \leq d(\mathbf{p}, \mathbf{x}) + d(\mathbf{x}, \mathbf{q})$ (triangle inequality).



k -Nearest Neighbor: *non*-Euclidian Space

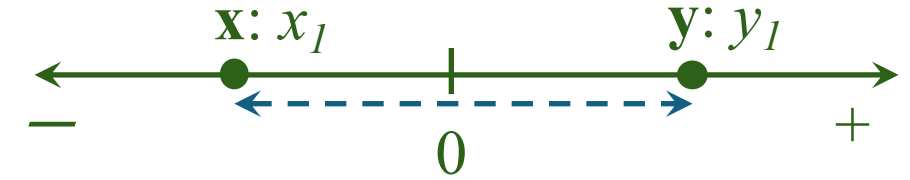
1. $d(\mathbf{p}, \mathbf{q}) \geq 0$ (no negative distances).
 2. $d(\mathbf{p}, \mathbf{q}) = 0 \iff_{\text{iff}} \mathbf{p} = \mathbf{q}$.
 3. $d(\mathbf{p}, \mathbf{q}) \neq d(\mathbf{q}, \mathbf{p})$ (symmetric).
 4. $d(\mathbf{p}, \mathbf{q}) \not\leq d(\mathbf{p}, \mathbf{x}) + d(\mathbf{x}, \mathbf{q})$ (triangle inequality).
- $\geq$



k -Nearest Neighbor: Euclidian Space

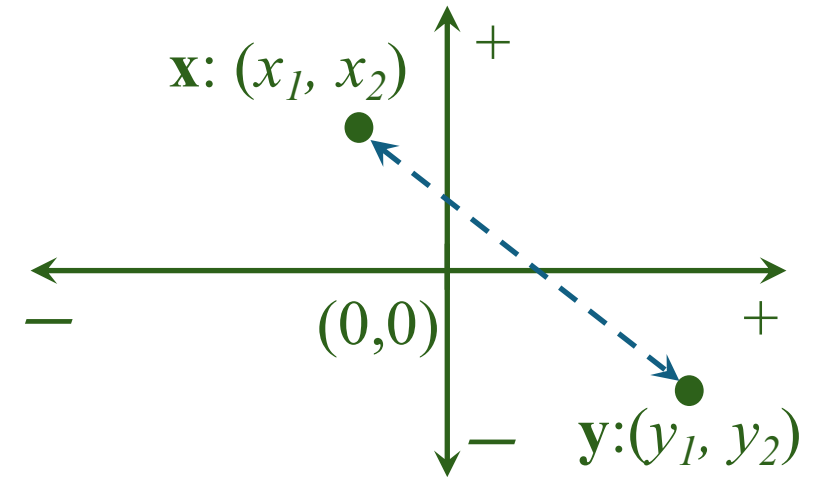
1 D Euclidian Distance $\mathbb{R}^1 \rightarrow \mathbb{R}$

$$\begin{aligned}d(x_1, y_1) &= [(x_1 - y_1)^2]^{1/2} \\&= [\sum_{i \in \{1\}} (x_i - y_i)^2]^{1/2} \\&= |x_1 - y_1|\end{aligned}$$



2D Euclidian Distance $\mathbb{R}^2 \rightarrow \mathbb{R}$

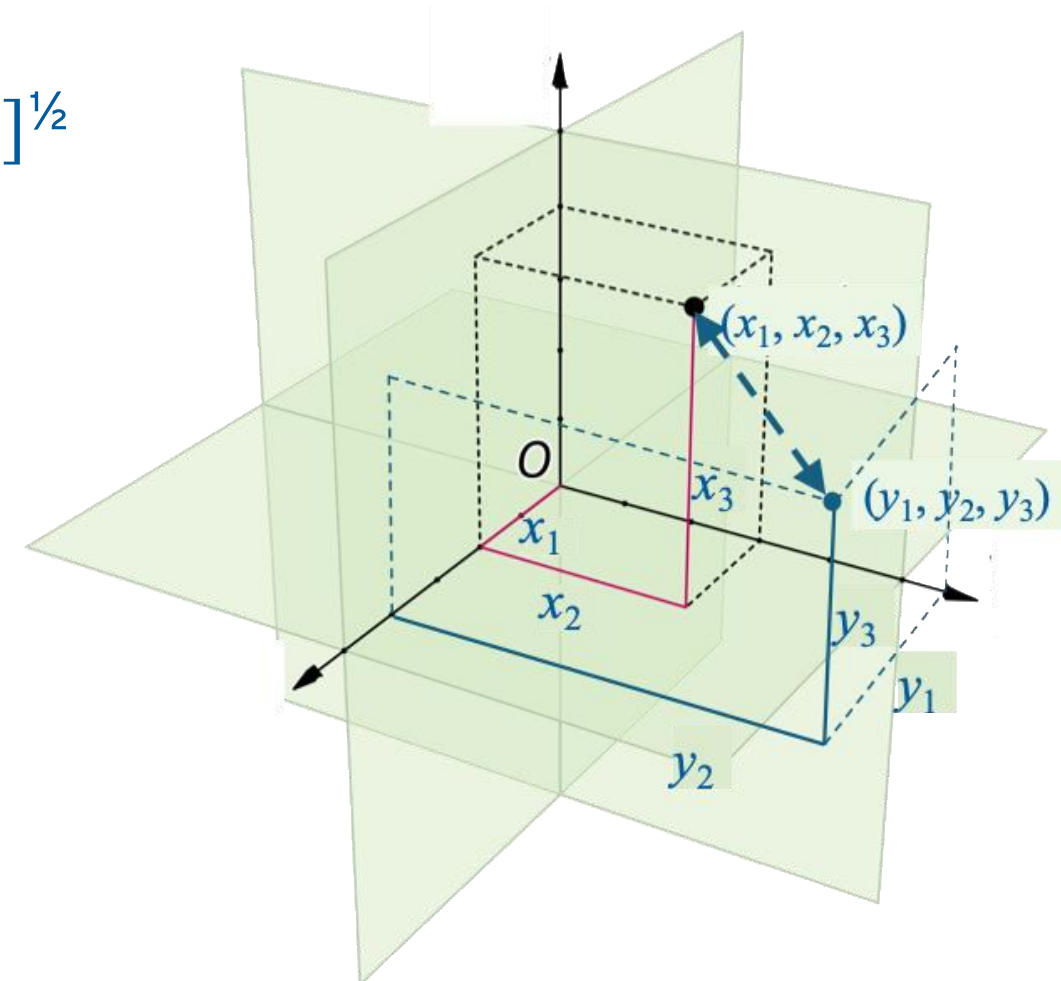
$$\begin{aligned}d(\mathbf{x}, \mathbf{y}) &= d((x_1, x_2), (y_1, y_2)) \\&= [(x_1 - y_1)^2 + (x_2 - y_2)^2]^{1/2} \\&= [\sum_{i \in \{1,2\}} (x_i - y_i)^2]^{1/2}\end{aligned}$$



k -Nearest Neighbor: Euclidian Space

3D Euclidian Distance $\mathbb{R}^3 \rightarrow \mathbb{R}$

$$\begin{aligned} d(\mathbf{x}, \mathbf{y}) &= d((x_1, x_2, x_3), (y_1, y_2, y_3)) \\ &= [(x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2]^{\frac{1}{2}} \\ &= [\sum_{i \in \{1,2,3\}} (x_i - y_i)^2]^{\frac{1}{2}} \end{aligned}$$



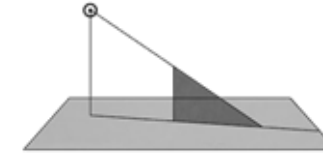
30

k-Nearest Neighbor: Euclidian Space

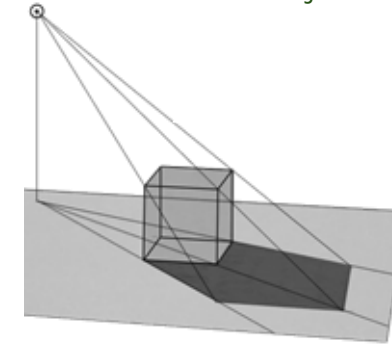
4D Euclidian Distance $\mathbb{R}^4 \rightarrow \mathbb{R}$

$$\begin{aligned} d(\mathbf{x}, \mathbf{y}) &= d((x_1, x_2, x_3, x_4), (y_1, y_2, y_3, y_4)) \\ &= [\sum_{i \in \{1,2,3,4\}} (x_i - y_i)^2]^{\frac{1}{2}} \end{aligned}$$

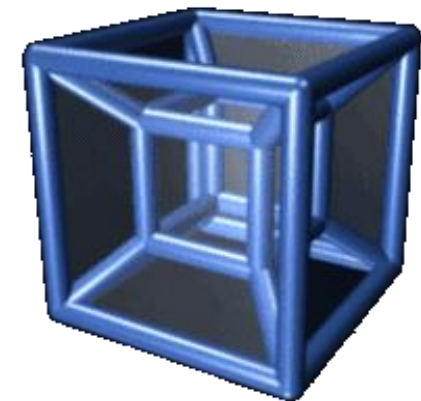
Shadow of a 2D object becomes 1D



Shadow of a 3D object becomes 2D



Shadow of a 4D object becomes 3D



A tesseract or 4-cube

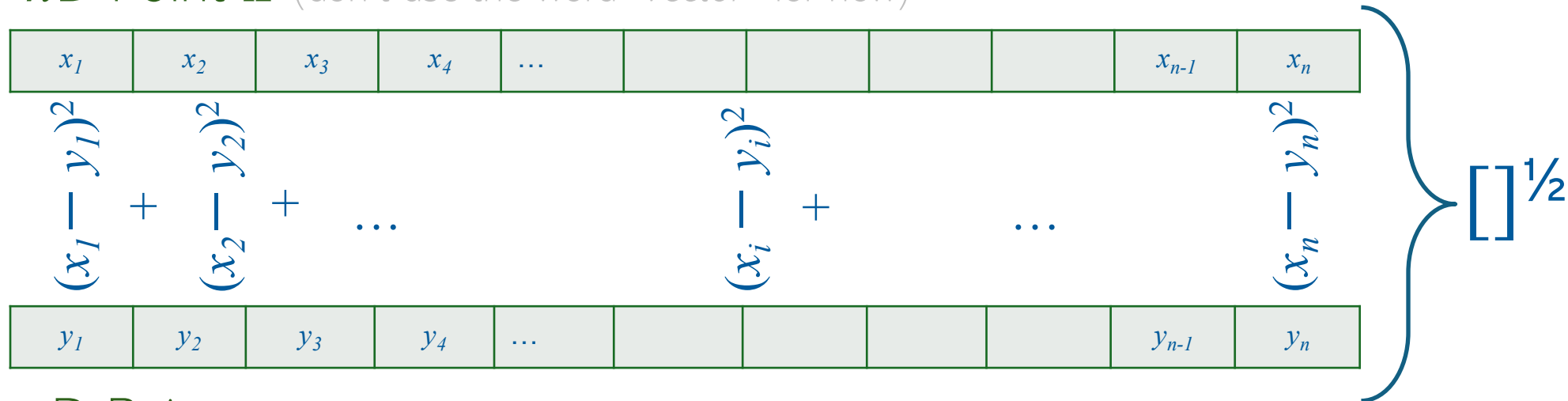
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k -Nearest Neighbor: Euclidian Space

n D Euclidian Distance $\mathbb{R}^n \rightarrow \mathbb{R}$

$$d(\mathbf{x}, \mathbf{y}) = d((x_1, \dots, x_n), (y_1, \dots, y_n)) = [\sum_{i \in \{1, \dots, n\}} (x_i - y_i)^2]^{1/2}$$

n D Point $\mathbf{x}$ (don't use the word "vector" for now)



n D Point $\mathbf{y}$ (don't use the word "vector" for now)

k -Nearest Neighbor: Minkowski Distance



Minkowski in 1883, at the time of being awarded the Mathematics Prize of the French Academy of Sciences

$$\left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p}$$

$p = 1$; Manhattan Distance
 $p = 2$; Euclidean Distance
 $p = +\infty$; Chebyshev Distance

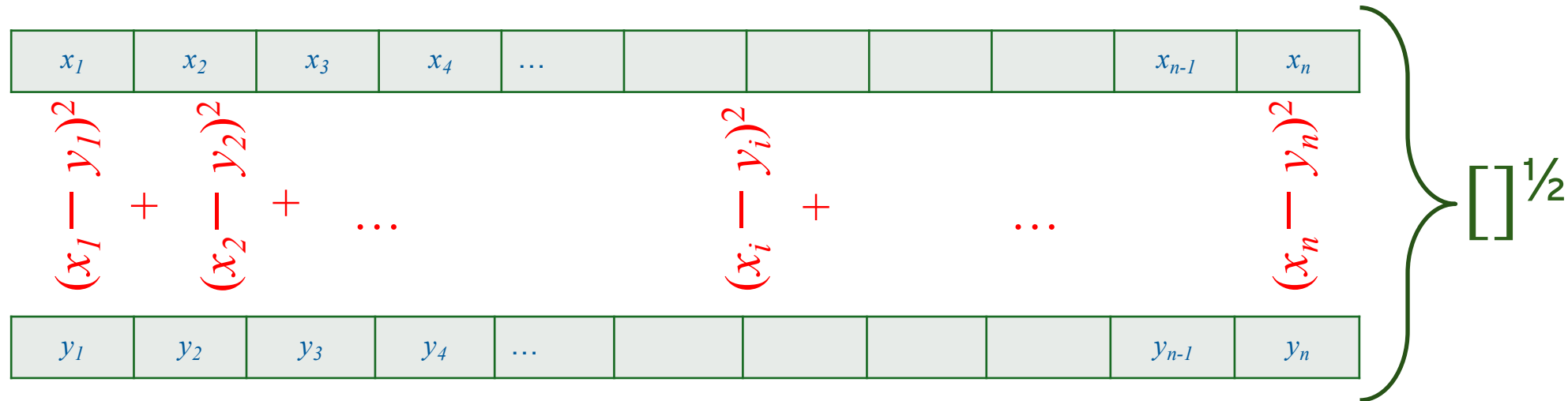
Hermann Minkowski (1864 – 1909)

Mathematician & professor at University of Königsberg, ETH Zürich, University of Göttingen
Best known for his foundational work on space-time 4D space which facilitated geometric interpretations of Einstein's special theory of relativity (1905).

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k -Nearest Neighbor: Curse of Dimensionality $\mathbb{R}^n \rightarrow +\infty$

$$\mathbb{R}^n \rightarrow \mathbb{R}: d(\mathbf{x}, \mathbf{y}) = [\sum_{i \in \{1, \dots, n\}} (x_i - y_i)^2]^{\frac{1}{2}}$$



$|\mathbf{O}^*|$ labeled observations $\rightarrow |\mathbf{O}^*|$ distances with unseen $\hat{o}$

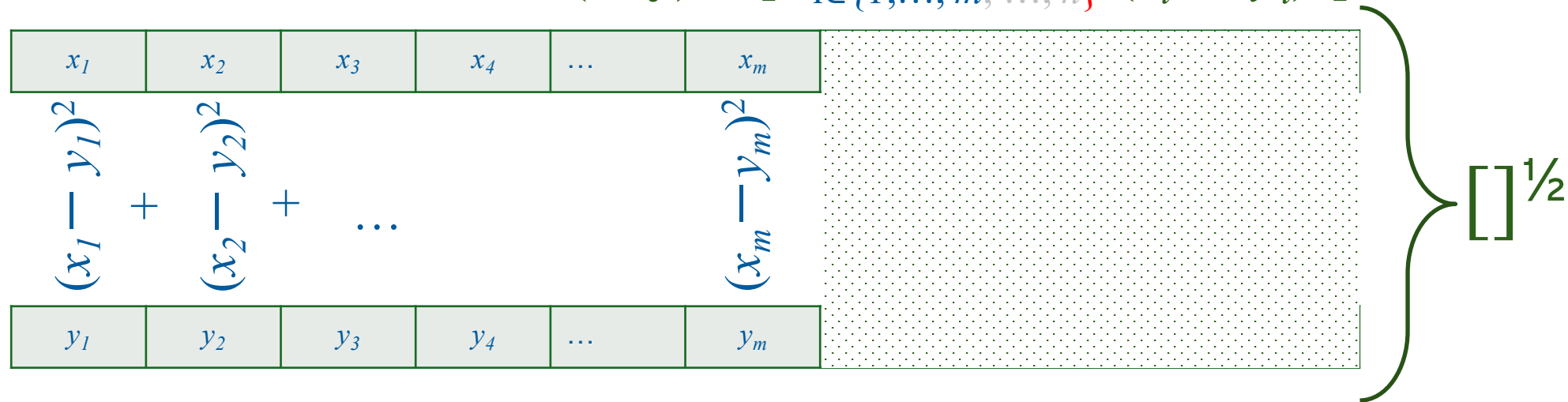
Each distance $\rightarrow n$ (subtractions + squares)

$\rightarrow n \times |\mathbf{O}^*| \rightarrow$ **Linearly grows** for both the size of training set and dimensions

k -Nearest Neighbor: Shadowing (Projection)

Naïve Dimensionality Reduction: Simply take the first few dims

$$\mathbb{R}^{n \gg 0} \rightarrow \mathbb{R}^{m \gg 0} \rightarrow \mathbb{R}: d(\mathbf{x}, \mathbf{y}) = [\sum_{i \in \{1, \dots, m, \dots, n\}} (x_i - y_i)^2]^{\frac{1}{2}}$$



$|\mathbf{O}^*|$ labeled observations $\rightarrow |\mathbf{O}^*|$ distances with unseen $\hat{o}$

Each distance $\rightarrow m$ (subtractions + squares)

$\rightarrow m \times |\mathbf{O}^*| \rightarrow \text{Manageable}$

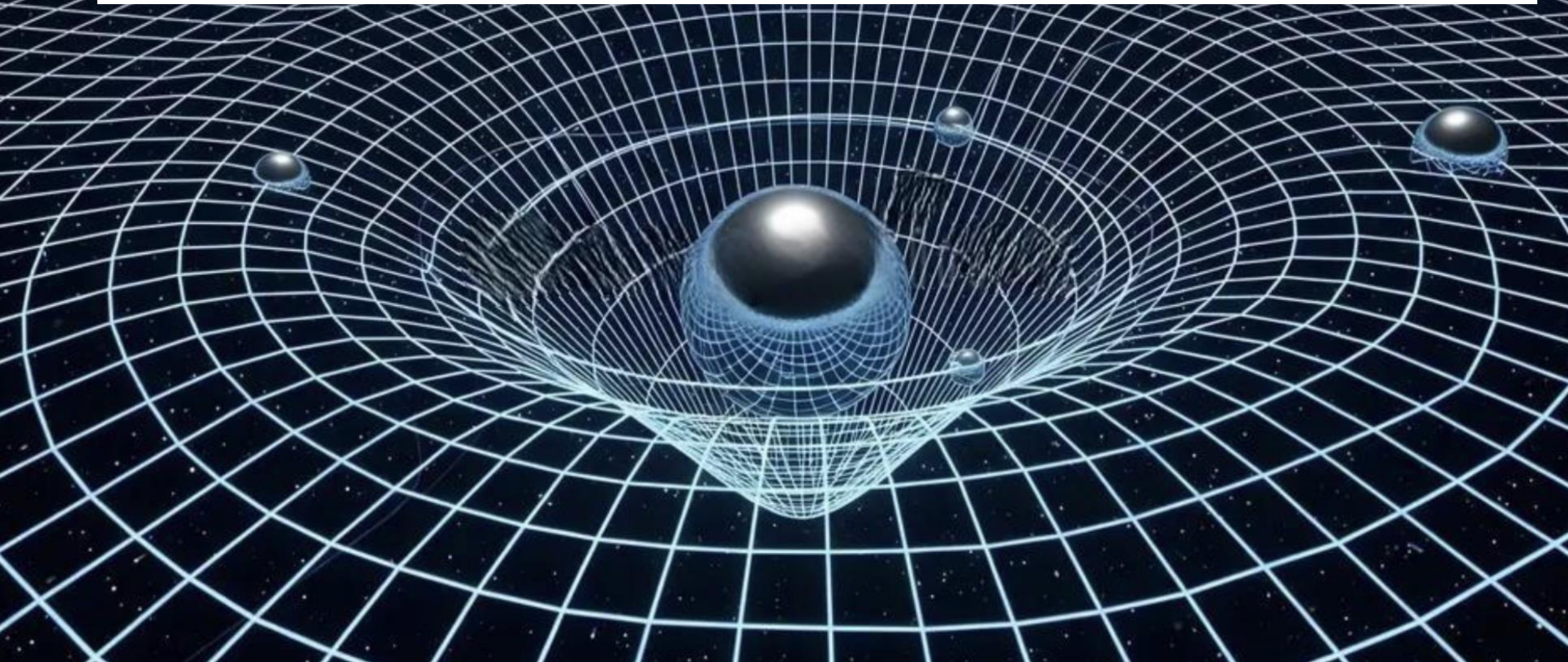


An image of the core region of a supermassive black hole, processed from a sparse array of radio telescopes
https://en.wikipedia.org/wiki/Black_hole

k -Nearest Neighbor: Locally-Sensitive Hashing (LSH)

Hashing Collisions for Efficient Nearest Neighbor Search → Almost Constant (1-3) Steps

CSC502: Systems for Massive Datasets, Mining of Massive Datasets: Ch03: Finding Similar Items



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k-Nearest Neighbor: Domain Challenges

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT} \{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k \mathbf{1} [c_i = c] \right\}$$


d, math “*kindly*” show us how to measure distances in Euclidian space.

The remaining task is ...

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k -Nearest Neighbor: Domain Challenges

How to map a real-world observation to a point in Euclidian space?

People

Animals (Birds, ...)

Objects (Houses, Cars, ...)

Image (Biometric Iris)

Audio (Voice, Music)

Video (Movies)

Text (Words, Sentence, Docs)

Timeseries

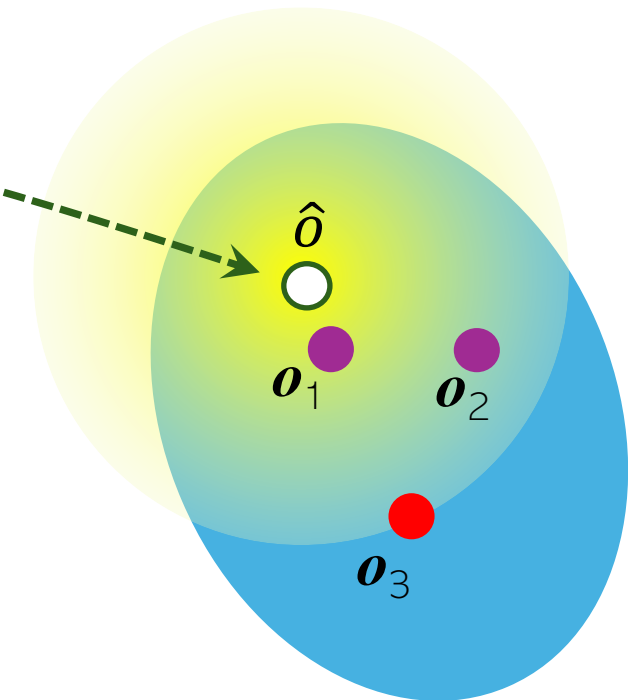
Genes

Spatial (GIS)

Light Bulbs

...

Encoding
Embedding
 $e(o)$



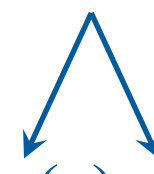
Euclidian Space
(Embedding Space)

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 k -Nearest Neighbor: Domain Challenges

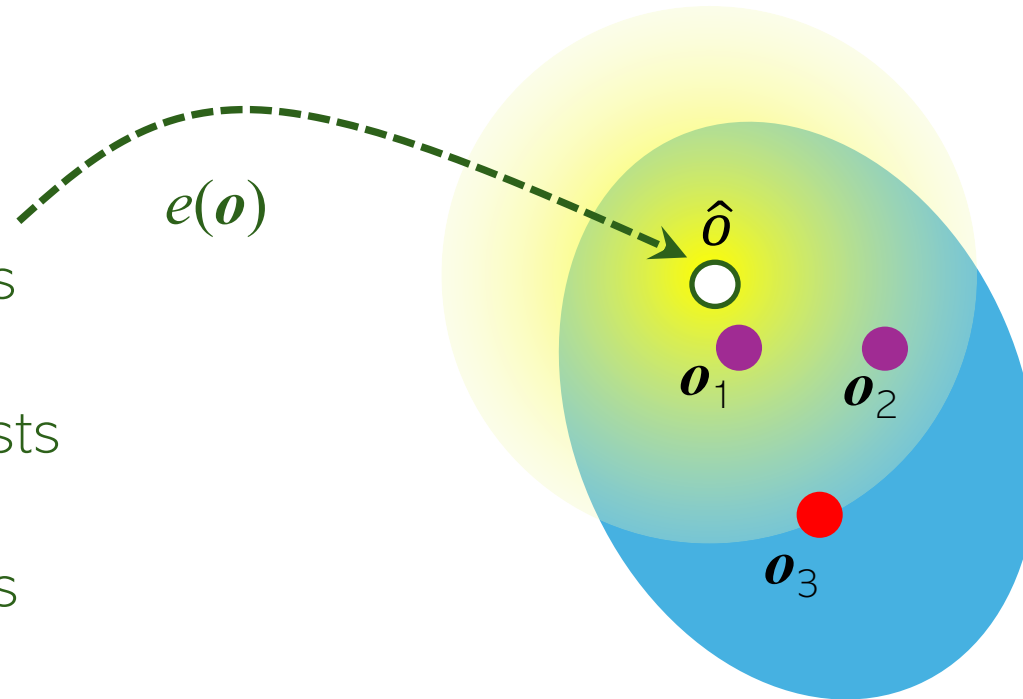
$$f(\hat{o} : \mathbf{O}^*) = \operatorname{ARGMAX} \left\{ \sum_{c_i \in \mathbf{C}} \sum_{\substack{(o,c) \in \operatorname{ARGSORT} \{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k 1 [c_i = c] \right\}$$


$$d(\hat{o}, o) \approx d(e(\hat{o}) \in \mathbb{R}^n, e(o) \in \mathbb{R}^n) = d(\mathbf{x}_{\hat{o}}, \mathbf{y}_o) = [\sum_{i \in \{1, \dots, n\}} (x_i - y_i)^2]^{1/2}$$


$$f(\hat{o} : \mathbf{O}^*) = \operatorname{ARGMAX} \left\{ \sum_{c_i \in \mathbf{C}} \sum_{\substack{(o,c) \in \operatorname{ARGSORT} \{d(e(\hat{o}), e(o))\} \\ (o, c) \in \mathbf{O}^*}}^k 1 [c_i = c] \right\}$$


40 k -Nearest Neighbor: Text

Words
Sentences
Docs
Social Posts
Books
Webpages
...



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k -Nearest Neighbor: Text: Word Space $\rightarrow$ Euclidian Space

Real-World Problem: Named Entity Annotator

Task: Classification $\mathbf{C} = \mathbf{Entities} = \{\text{Person, Organization, Event, Date, Location}\}$

Approach: Fully Supervised $\mathbf{O}^* = \{(w, c)\}$, $\mathbf{U}^* = \{(\hat{w}, f(\hat{w}))\}$

Technique: $f: k\text{-NN} \rightarrow d(\hat{w}, w) \approx d(e(\hat{w}), e(w)) \rightarrow e: \mathbf{U} = \mathbf{Vocabs} \rightarrow \mathbb{R}^{n=1}$

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k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=1}$

Indexing $\rightarrow$ Enumerate words in a language's dictionary (vocabulary set)



$$e(\hat{w}: \text{'aardvark'})=0$$



$$e(w: \text{'apple'})=1050$$



$$e(w: \text{'zyzzyva'})=|V|$$

Does $e(.)$ preserve “*real-world*” proximity to “*spatial*” proximity? **No!**

$$\begin{aligned} |e(\text{'aardvark'}) - e(\text{'apple'})| &< |e(\text{'aardvark'}) - e(\text{'aardvark'})| \\ |0 - 1050| &< |0 - 520,000| \end{aligned}$$

'aardvark' is closer to 'apple' vs. 'zyzzyva'!

'aardvark' is tech company!

Distributional Hypothesis

Words that occur in similar contexts tend to have similar meanings.
Meaning difference corresponds to difference in environments.

Joos, M. (1950). Description of language design. JASA, 22, 701–708.
Harris, Z. S. (1954). Distributional structure. Word, 10, 146–162.
Firth, J. R. (1957). A synopsis of linguistic theory 1930–1955.

Dictionary

Definitions from [Oxford Languages](#) · [Learn more](#)



col·lo·ca·tion

noun

1. **LINGUISTICS**
the habitual juxtaposition of a particular word with another word or words with a frequency greater than chance.
"the words have a similar range of collocation"
2. the action of placing things side by side or in position.
"the collocation of the two pieces"

Vector Semantics

- Distribution (co-occurrences)
 - Count-based (Sparse Vector Representations)
 - Term-Doc Matrix
 - Tokens in Document Space
 - Documents in Token Space
 - Term-Term Matrix
 - Tokens in Token Space (co-occurrence context)
 - Document in Average of Constituent Tokens
 - Weighted Term-Term Matrix
 - Tf-Idf
 - Learning-based (Dense Vector Representations)
 - Matrix Factorization (Decomposition)
 - Classification (Word2Vec)

Word Vector
Word Embedding
Word Point

Meaning Similarity



Geometric Proximity

Context Window

Term-Document Matrix

Context Window = Document

Term-Document Matrix

Context Window = Document

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

Figure 6.2 The term-document matrix for four words in four Shakespeare plays. Each cell contains the number of times the (row) word occurs in the (column) document.

$$|\text{Vocabs}| \times |\text{Documents}|$$

Term-Document Matrix

Context Window = Document

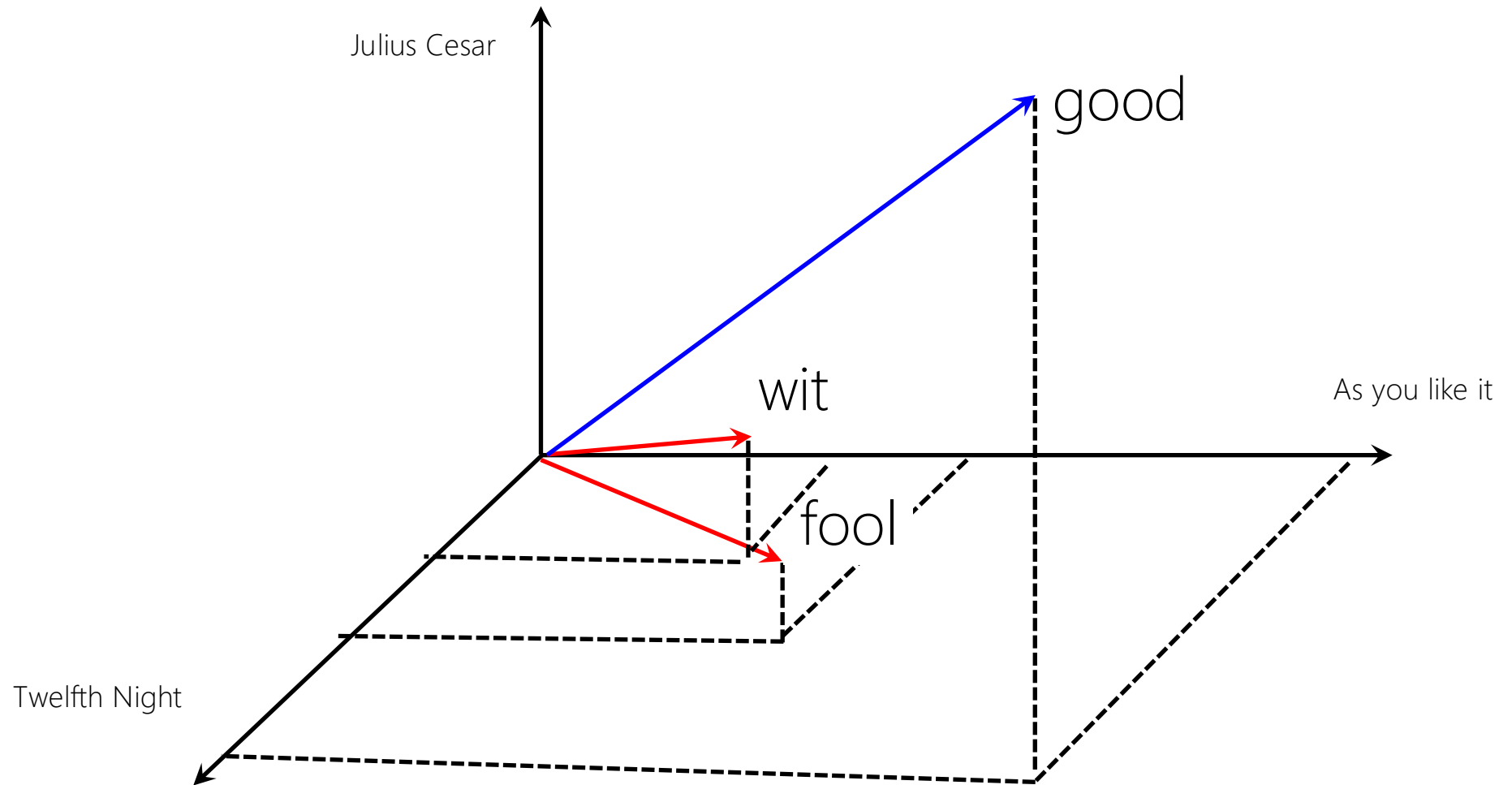
	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

Figure 6.2 The term-document matrix for four words in four Shakespeare plays. Each cell contains the number of times the (row) word occurs in the (column) document.

$$|\text{Vocabs}| \times |\text{Documents}|$$

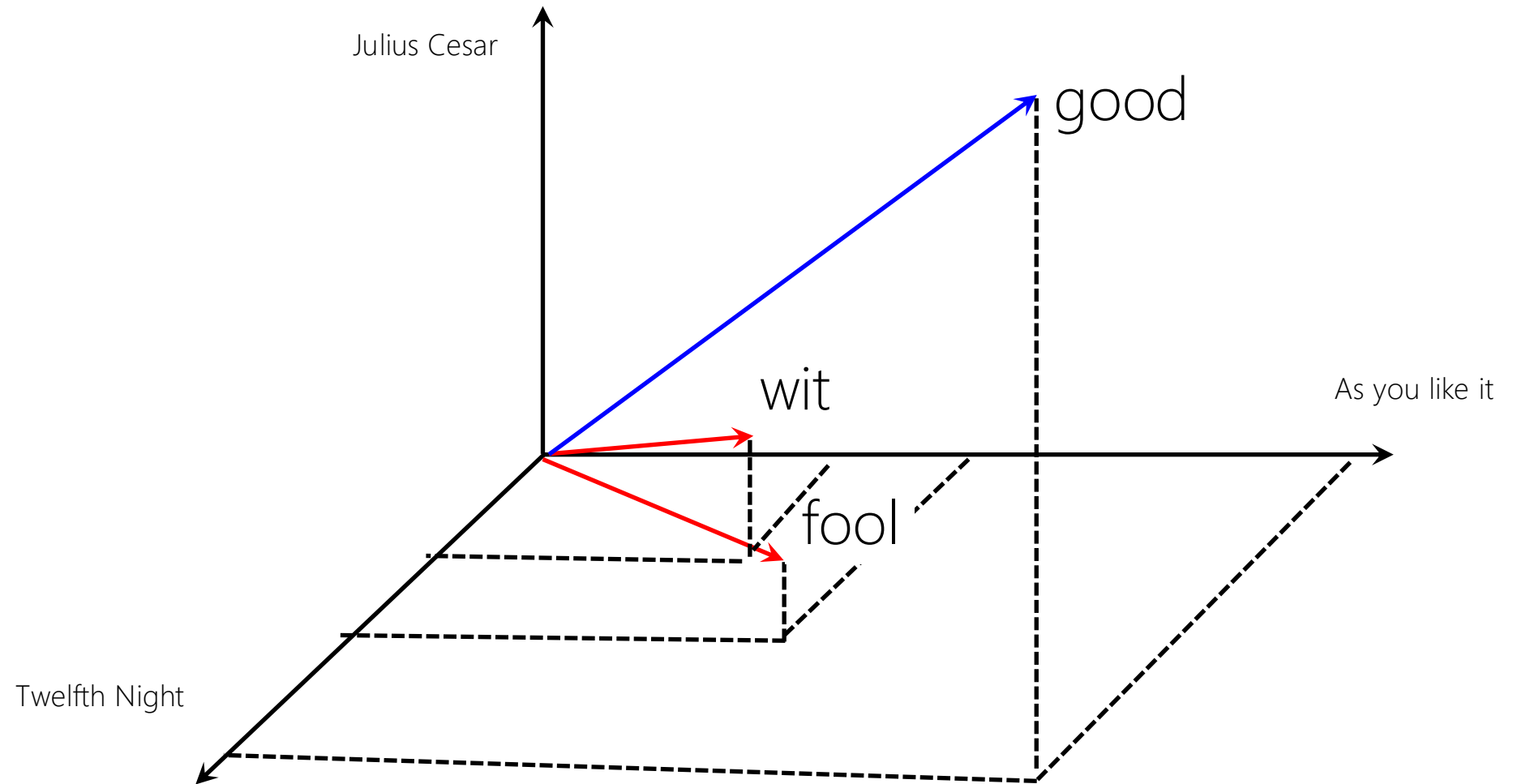
Term-Document Matrix

Words are points (vectors) in document space!



Term-Document Matrix

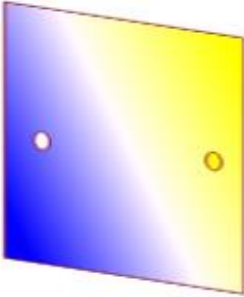
Distribution of words in documents!



Logistic Regression

Optimization: $\text{Min } -\sum_{(x,y) \in D} \text{Log } P(y|x_y) \rightarrow \begin{cases} y = 1: P(+|x_+) = \text{Sigmoid}(f(x_+)) \\ y = 0: P(-|x_-) = 1 - \text{Sigmoid}(f(x_-)) \end{cases}$

- Function f is linear function of weights


$$[x_0 \ x_1 \ \dots \ x_d \ 1] \begin{bmatrix} w_1 \\ w_2 \\ \dots \\ w_{d+1} \end{bmatrix} = [?]$$

A red squiggly arrow points from the vector $[x_0 \ x_1 \ \dots \ x_d \ 1]$ to the matrix $\begin{bmatrix} w_1 \\ w_2 \\ \dots \\ w_{d+1} \end{bmatrix}$.

Word Vector Representation!

Term-Document Matrix

Distribution of documents in words.

Documents in word space!

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

Figure 6.2 The term-document matrix for four words in four Shakespeare plays. Each cell contains the number of times the (row) word occurs in the (column) document.

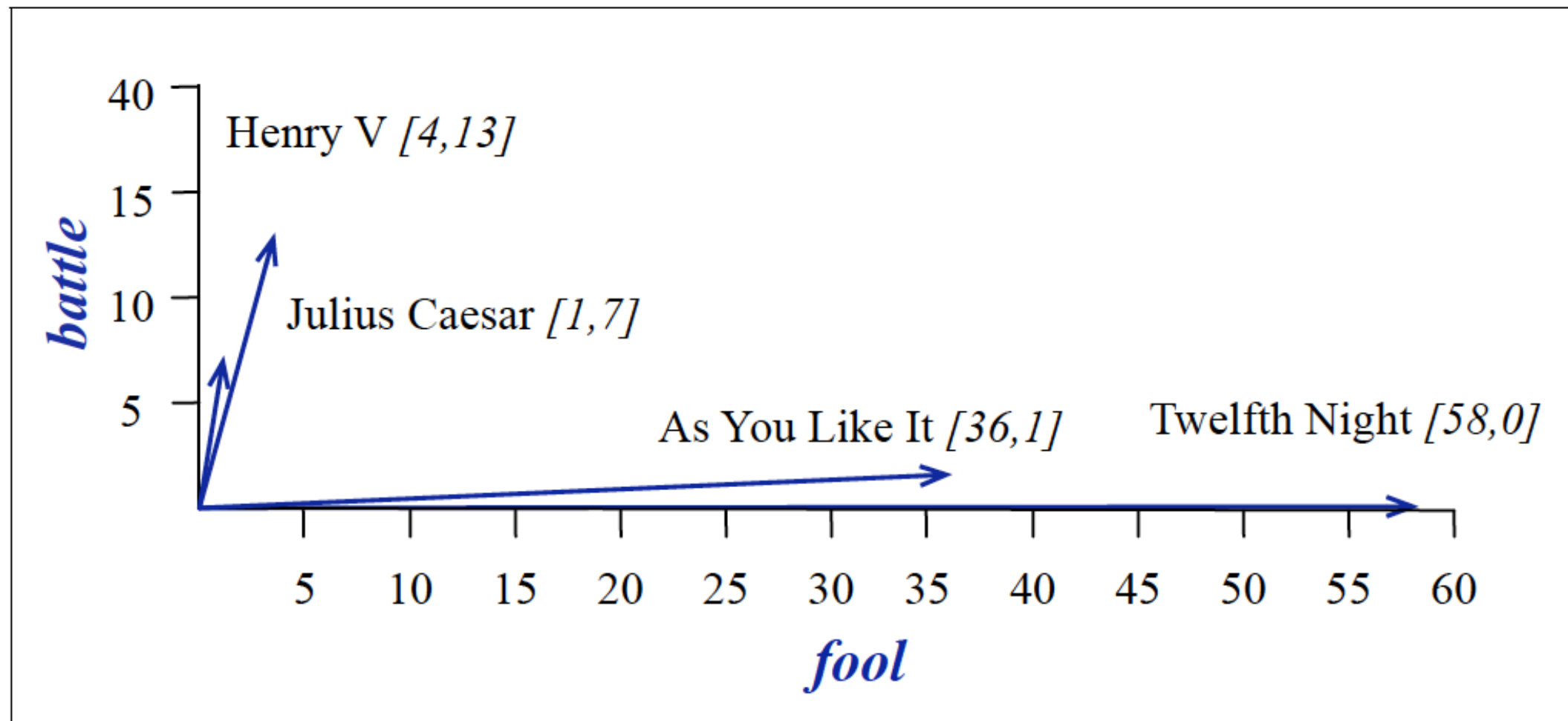
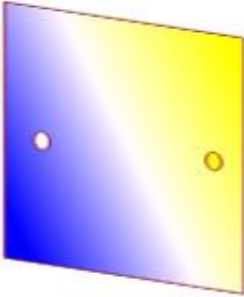


Figure 6.4 A spatial visualization of the document vectors for the four Shakespeare play documents, showing just two of the dimensions, corresponding to the words *battle* and *fool*. The comedies have high values for the *fool* dimension and low values for the *battle* dimension.

Logistic Regression

Optimization: $\text{Min } -\sum_{(x,y) \in D} \text{Log } P(y|x_y) \rightarrow \begin{cases} y = 1: P(+|x_+) = \text{Sigmoid}(f(x_+)) \\ y = 0: P(-|x_-) = 1 - \text{Sigmoid}(f(x_-)) \end{cases}$

- Function f is linear function of weights


$$[x_0 \ x_1 \ \dots \ x_d \ 1] \begin{bmatrix} w_1 \\ w_2 \\ \dots \\ w_{d+1} \end{bmatrix} = [?]$$

A red squiggly arrow points from the x_0 term in the vector to the w_1 term in the weight vector.

Document Vector Representation!

Term-Document Matrix

Context Window = Document

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

Figure 6.2 The term-document matrix for four words in four Shakespeare plays. Each cell contains the number of times the (row) word occurs in the (column) document.

Is it sparse or dense matrix? Why?

1-hot Encoding (Revisited)

$$w_i = [0 \ 0 \ 0 \ \dots \ 0 \ 1 \ \dots \ 0 \ 0]_{|V|}$$

Context Window = ± 0 tokens = 1-gram

One Extreme: orthogonal (± 0 tokens)

Another Extreme: all related ($\pm \infty$ tokens)

Term-Term Matrix

aka word-word matrix
aka word-context matrix

Context Window =

- Fixed (Window): $\pm n$ tokens = *unordered* n-gram
- Dynamic: Document, Paragraph, Sentence, Tweet

	aardvark	...	computer	data	result	pie	sugar	...
cherry	0	...	2	8	9	442	25	
strawberry	0	...	0	0	1	60	19	
digital	0	...	1670	1683	85	5	4	
information	0	...	3325	3982	378	5	13	

Figure 6.5 Co-occurrence vectors for four words in the Wikipedia corpus, showing six of the dimensions (hand-picked for pedagogical purposes). The vector for *digital* is outlined in red. Note that a real vector would have vastly more dimensions and thus be much sparser.

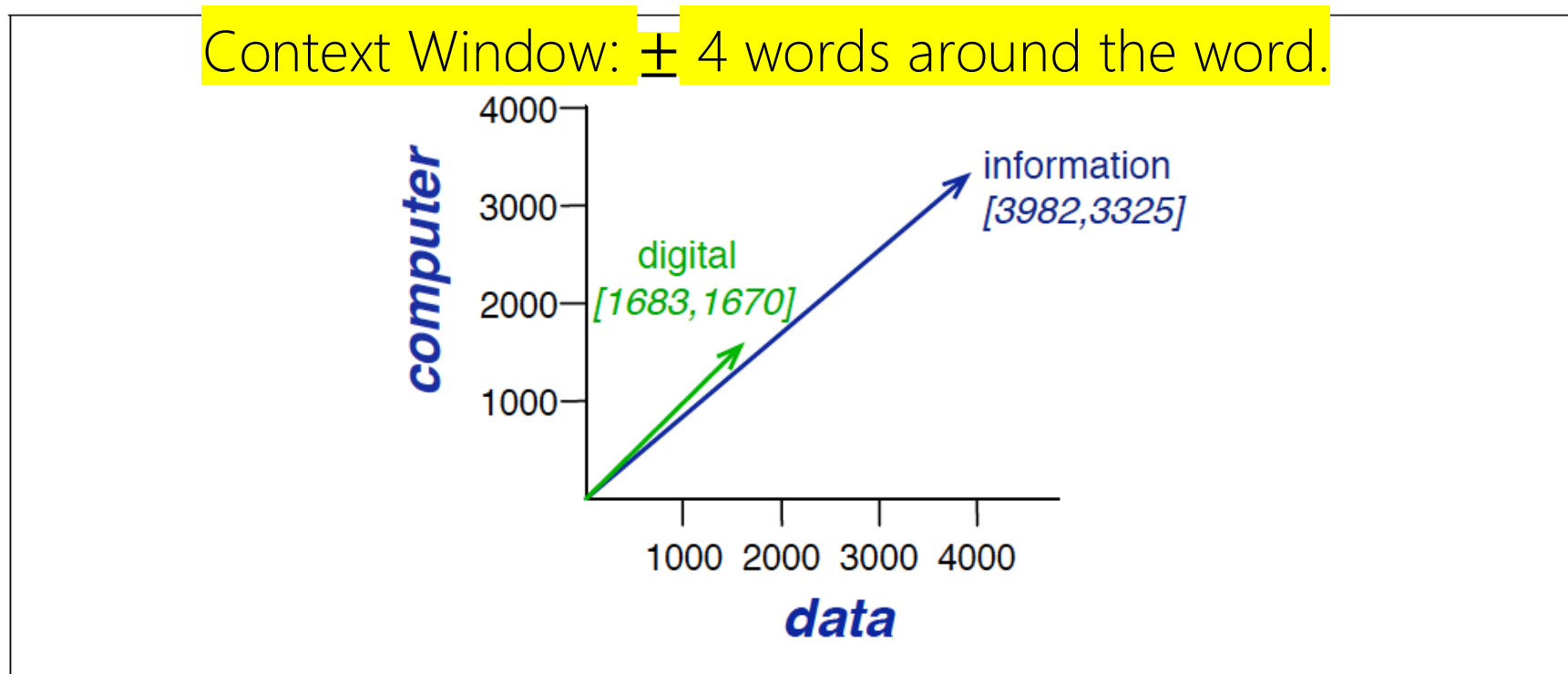


Figure 6.6 A spatial visualization of word vectors for *digital* and *information*, showing just two of the dimensions, corresponding to the words *data* and *computer*.

Term-Term Matrix

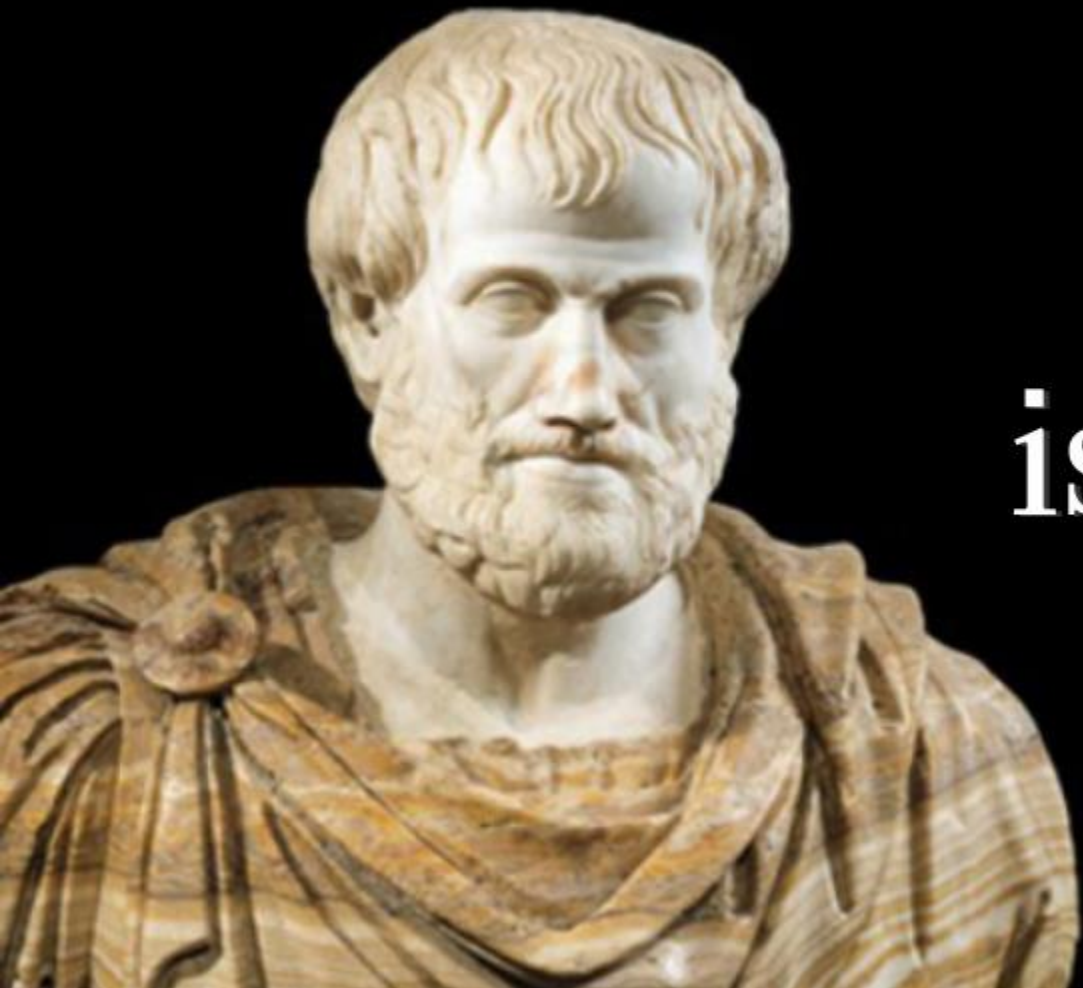
Matrix Size = $|\text{Vocabs}|^2$

Is it sparse or dense? Why? Does context size matter?

Term-Term Matrix

Context Window = ± 1

Term-Document Matrix $\rightarrow$ Term-Term Matrix



A friend to all
is a friend to none.

- Aristotle

TF-IDF

Weighted Term-Document Matrix

Context Window = Document
 $|Vocabs| \times |Documents|$

Term Frequency = TF = Term-Document
Matrix

Term-Document Matrix

Context Window = Document

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

Figure 6.2 The term-document matrix for four words in four Shakespeare plays. Each cell contains the number of times the (row) word occurs in the (column) document.

Term-Document Matrix

Non-Discriminative Words

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

Figure 6.2 The term-document matrix for four words in four Shakespeare plays. Each cell contains the number of times the (row) word occurs in the (column) document.

iDF = inverse Document Frequency

$$\text{idf}_t = \left(\frac{N}{\text{df}_t} \right) \rightarrow \log_{10} \left(\frac{N}{\text{df}_t} \right)$$

Word	df	idf
Romeo	1	1.57
fool	36	0.012
good	37	0

where N is the total number of documents in the collection

TF-IDF

$$w_{t,d} = \text{tf}_{t,d} \times \text{idf}_t$$

$$\text{tf}_{t,d} = \log_{10}(\text{count}(t,d) + 1)$$

$$\text{idf}_t = \log_{10} \left(\frac{N}{\text{df}_t} \right)$$

TF-iDF

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	0.074	0	0.22	0.28
good	0	0	0	0
fool	0.019	0.021	0.0036	0.0083
wit	0.049	0.044	0.018	0.022

Figure 6.8 A tf-idf weighted term-document matrix for four words in four Shakespeare plays, using the counts in Fig. 6.2. For example the 0.049 value for *wit* in *As You Like It* is the product of $tf = \log_{10}(20 + 1) = 1.322$ and $idf = .037$. Note that the idf weighting has eliminated the importance of the ubiquitous word *good* and vastly reduced the impact of the almost-ubiquitous word *fool*.

Semantic Distance

Integer Space vs. Probability Space vs. Vector Space

Index: $d(\text{aardvark}, \text{apple}) = |0 - 1| = 1 \rightarrow$ no semantic proximity preserving

1-Hot?

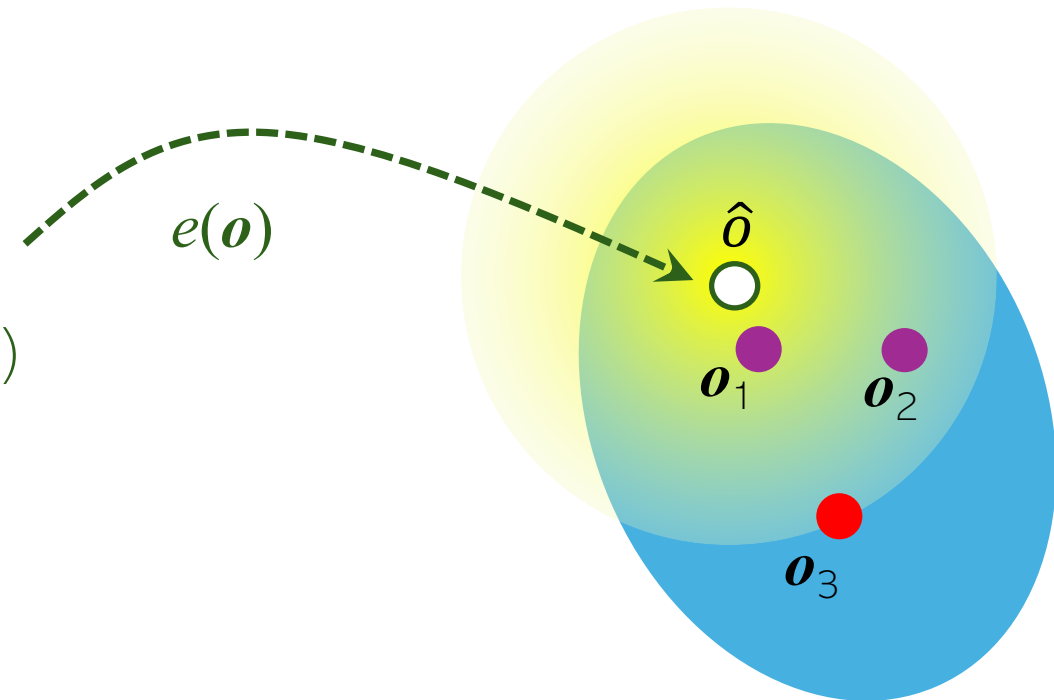
Term-Doc Matrix?

Term-Term Matrix?

TF-IDF?

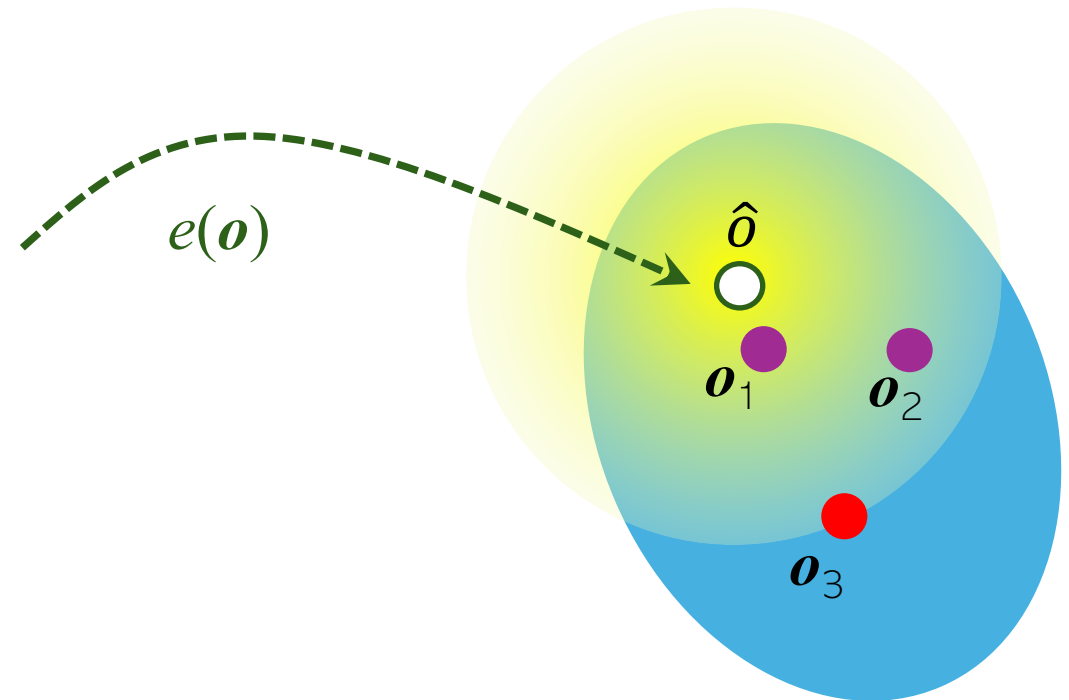
81 k -Nearest Neighbor: Image

Image (Biometric Iris)



82 k -Nearest Neighbor: Audio

Audio (Voice, Music)



83

k -Nearest Neighbor: People?

Hands-Crafted Feature Extraction (Height, Weight, ...)

People (Fingerprints)

